Psychological statistics

Homework Assignment ②

B812109041 Isaac, Lee

1. SOCCER GAMES

a. Chi-square test for independence (two-variable)

i. H0: People from Sage or Sussex and Score variables are independent.

ii. H1: People from Sage or Sussex and Score variables are associated.

b. Contingency table

Score/Job	Sage Publications	University of Sussex
Yes	5	23
no	19	30

Shooting success rate

- Sage: 20.8%
- Sussex: 43.48%

c. Chi-square Test

```
> chisq_test(SOCCER)
# A tibble: 1 × 6
      n statistic      p    df method          p.signif
* <dbl>     <dbl>  <dbl> <int> <chr>           <chr>
1    77      2.72 0.0988     1 Chi-square test ns
```

d. Conclusion

- People from Sage or Sussex and Score variables are not associated. (p = 0.0988, cannot reject H0)
- Based on the results of the chi-square test, we cannot conclude from this data that "We predict that Sussex people will score more than Sage people."

2. ONE-HANDED CYCLISTS

a. Chi-square test for independence (two-variable)

i. H0: Nationality of People and the way of cycling are independent.

ii. H1: Nationality of People and the way of cycling are associated.

b. Contingency table

Hand / Nationality	English	Dutch
One Handed	17	120
Two Handed	154	578

The proportion of cyclists riding with one hand:

- English: 9.94%
- Dutch: 17.19%

c. Chi-square Test

```
> chisq_test(CYCLISTS)
# A tibble: 1 × 6
      n statistic      p    df method          p.signif
* <dbl>     <dbl>  <dbl> <int> <chr>           <chr>
1   869      4.90 0.0268     1 Chi-square test *
>
```

d. Conclusion

- Nationality of People and the way of cycling are not independent. (p = 0.0268, reject H0)
- Based on the results of the chi-square test, it is evident that nationality and cycling habits are correlated variables. Therefore, it is possible to determine from the contingency table data whether it is the English or the Dutch who are more likely to ride bicycles with one hand.
- English (9.94%) < Dutch (17.19%), Dutch people have a higher proportion of cycling with one hand.

3. DARK LOAD

1) What is the hypothesis to be tested?

Whether "backward masked messages" have a significant difference in their impact on butchers.

2) Analyze the data with a Wilcoxon signed-rank test and report the effect size.

a. Set up Hypothesis.

- H0: The impact of "backward masked messages" on butchers has not shown a significant difference.
- H1: The impact of "backward masked messages" on butchers has shown a significant difference.

b. Repeated-measure Wilcoxon test

```
        Wilcoxon signed rank test with continuity correction

data:  data$nomessag and data$message
V = 294.5, p-value = 0.0373
alternative hypothesis: true location shift is not equal to 0
```

c. Effect size.

```
> ### [ Step.3 ] Effect size
> rFromWilcox<-function(wilcoxModel, N){
+   z<- qnorm(wilcoxModel$p.value/2)
+   r<- z/ sqrt(N)
+   cat(Model$data.name, "Effect Size, r = ", r)
+ }
> rFromWilcox(Model, 32)
data$nomessag and data$message Effect Size, r =  -0.3681225
```

d. Conclusion

- According to the Wilcoxon signed-rank test, with a p-value of 0.0373 < 0.05, we can reject the null hypothesis (H0). This indicates that there is a significant difference in the impact of "backward masked messages".

3) **Would the results be different using t-test?**

```
        Paired t-test

data:  data$nomessag and data$message
t = 2.7641, df = 31, p-value = 0.009521
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.6144031 4.0730969
sample estimates:
mean difference
        2.34375
```

- The p-value of the t-test is 0.009521, which is significantly smaller than that of the Wilcoxon signed-rank test (p = 0.0373).

4. CHECK THE PAPER

- **Please identify the statements of research question, hypothesis, and statistic testing methods.**
  - **Research Question**: The purpose of this study was to identify cortical and subcortical substrates related to respiratory sensory gating elicited by paired inspiratory occlusions.
  - **Hypothesis:** The ratio of effect size in the paired- versus single-occlusion condition (β paired /β single) would be less than 2.
  - **Statistic Testing Methods:** Wilcoxon Signed - Rank test.

- **Please reproduce the results using the attached dataset in R. Report your own decision following the testing.**

```
> Supra_Marginal_B

        Wilcoxon rank sum test with continuity correction

data:  data$Supra_Marginal_B and data$Expect
W = 276, p-value = 2.465e-10
alternative hypothesis: true location shift is less than 0
```

```
> Basal_Ganglia_B

        Wilcoxon rank sum test with continuity correction

data:  data$Basal_Ganglia_B and data$Expect
W = 552, p-value = 2.028e-05
alternative hypothesis: true location shift is less than 0
```

```
> Middle_Frontal_L

        Wilcoxon rank sum test with continuity correction

data:  data$Middle_Frontal_L and data$Expect
W = 552, p-value = 2.028e-05
alternative hypothesis: true location shift is less than 0
```

```
> Thalamus_B

        Wilcoxon rank sum test with continuity correction

data:  data$Thalamus_B and data$Expect
W = 506, p-value = 2.903e-05
alternative hypothesis: true location shift is less than 0
```

```
> Superior_Temporal_B

        Wilcoxon rank sum test with continuity correction

data:  data$Superior_Temporal_B and data$Expect
W = 598, p-value = 0.001245
alternative hypothesis: true location shift is less than 0
```

```
> Precentral_B

        Wilcoxon rank sum test with continuity correction

data:  data$Precentral_B and data$Expect
W = 506, p-value = 2.903e-05
alternative hypothesis: true location shift is less than 0
```

Conclusion

According to the Wilcoxon signed-rank test, all p-values of the results are far less than 0.05. Therefore, it can be concluded that a beta ratio less than 2 exhibits a significant difference.